LAB EXPERIMENT

ITA0506-COMPUTER VISION

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EXPNO:13

Aim

To perform perspective transformation on a video using OpenCV in Python by converting a selected quadrilateral region into a rectangular view.

Algorithm

1. Import the required libraries: cv2 and numpy.
2. Open the input video using cv2.VideoCapture().
3. Read each frame from the video.
4. Select four points from the original frame.
5. Define four corresponding points for the output rectangle.
6. Calculate the perspective transformation matrix using cv2.getPerspectiveTransform().
7. Apply the transformation using cv2.warpPerspective().
8. Display the transformed video frame.
9. Press q to stop the video.
10. Release the video and close all windows.

CODE:

EXP-13.py - C:/Users/POORNA SKHT/OneDrive/Tài liệu/computer vision ita0502/lab experim...

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```python
import cv2
import numpy as np

# Open the video
cap = cv2.VideoCapture(r"C:\Users\POORNA SKHT\OneDrive\Tài liệu\computer vision

if not cap.isOpened():
    print("Error: Cannot open video")
    exit()

while True:
    ret, frame = cap.read()

    if not ret:
        print("Video ended")
        break

    # Get frame dimensions
    height, width = frame.shape[:2]

    # Four points from the original video frame
    # Change these points according to your video
    pts1 = np.float32([
        [100, 100],
        [width - 100, 100],
        [width - 100, height - 100],
        [100, height - 100]
    ])

    # Four points for the output rectangle
    pts2 = np.float32([
        [0, 0],
        [width, 0],
        [width, height],
        [0, height]
    ])

    # Calculate perspective transformation matrix
    matrix = cv2.getPerspectiveTransform(pts1, pts2)
```

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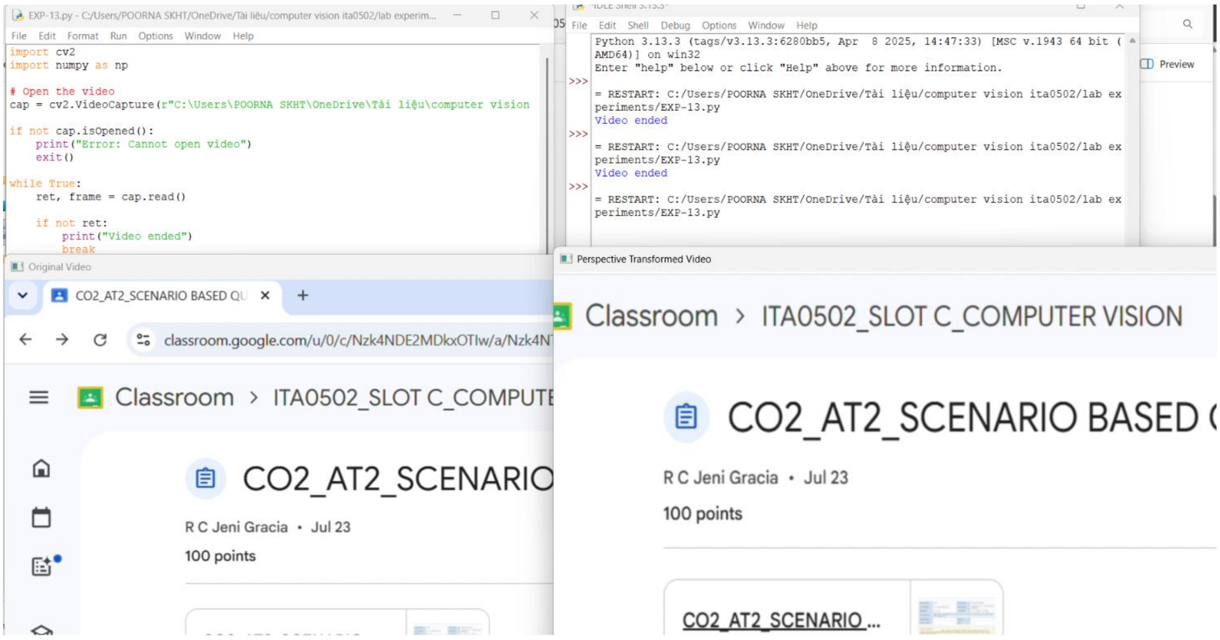
OUTPUT :



Result:

The perspective transformation was successfully performed on the video. The selected region of each video frame was transformed into a rectangular view using the perspective transformation matrix.